Carlson Zheng

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Education

University of Alberta — Edmonton, AB
BSc Computer Engineering Co-op — *2024 - 2029*

Skills

Programming: Python, C/C++, Java, Bash, VHDL, Assembly
Frameworks/Libraries: Flask, OpenCV, Tensorflow, Pandas, scikit-learn, numpy, mediapipe
Developer Tools: Git, STM32, Arduino, ESP32, KiCad, Linux, Visual Studio, Analog Discovery 3, SolidWorks

Experience

Control Systems Team Member — Dec 2025 - Present
Alberta Bionix — *Edmonton, AB*

- Designing a control system and integrating a complex sensor network for a lower-limb prosthetic to model a dynamic walking cycle using an STM32, moteous N1, rotary encoders, and IMUs.
- Interfaced rotary encoders and IMUs using embedded C, overcoming hardware abstraction challenges and leveraging hardware-accelerated data acquisition, reducing CPU load by 30%.
- Validated signal reliability across a network of 8+ interconnected sensors and components through 50+ hours of bench testing and hardware simulations.

Vice-President of Communications — Oct 2022 - April 2023
Munch-It / Junior Achievement — *Calgary, AB*

- Spearheaded a student-run company in the production of over 500+ units and achieving over $1265+ in revenue and securing the Business Plan of the Year award out of the 30+ competing companies.
- Oversaw content development across social media and digital marketing platforms, while collaborating with the I.T. department to design and implement a website featuring intuitive and user-friendly interfaces.

Projects

Splitsies | *Git, Python, JavaScript, Flask, PostgreSQL, Render*

- Created an end-to-end expense tracking application leveraging AI-assisted tools to rapidly prototype and integrate a fullstack environment using Javascript and Python.
- Implemented a PostgreSQL database on Render to securely model relationships between users, trips, and ledger entries while ensuring strict data integrity.

Domain Detector | *Git, Python, Linux, OpenCV, Tensorflow*

- Developed a real-time gesture recognition system in Python that accurately identifies specific hand signs by tracking 21 skeletal landmarks through a standard webcam.
- Trained a lightweight machine learning model using Tensorflow and Pandas that achieves over 93% accuracy and optimized to run smoothly on standard computer systems.

Discord Automation Bot | *Python, Flask, Render, Git, Discord*

- Created using Python and Discord API to automate 500+ monthly scheduled and location-triggered messages.
- Deployed to Render cloud hosting with a Flask 'heartbeat' monitoring system, ensuring 99.9% uptime.
- Implemented custom API endpoints for real-time management, reducing manual configuration time by 30% for pause/resume and location update features.

Certifications

Machine Learning Specialization — Completed August 2026
Issued by Coursera, Stanford Online, DeepLearning.AI

Supervised Machine Learning: Regression and Classification

- Linear Regression, Logistic Regression, Classification, Supervised Learning, Scikit-Learn, numpy

Advanced Learning Algorithms

- Neural Networks, Regularization, Overfitting, Feature Engineering, Decision Trees, TensorFlow

Unsupervised Learning, Recommenders, Reinforcement Learning

- Reinforcement Learning, Clustering, Anomaly Detection, Content-Based Filtering